PART 14 — RISK REGISTER & RAID GOVERNANCE LOG

Project Ref: RAID-BLYNT-2026

Management Framework: Continuous Agile RAID Assessment (Risks, Assumptions, Issues, Dependencies)

Risk Scoring Formula: $\text{Risk Score} = \text{Probability (1–5)} \times \text{Impact (1–5)}$

1. Risk Register (Critical Vulnerabilities & Mitigations)

Risk ID	Risk Description	Prob (1-5)	Imp (1-5)	Score (1-25)	Non-Technical Rationale & Engineering Mitigation	Owner	Status
RSK-01	**Floating-point mathematical rounding errors** causing general ledger imbalance in tax filings.	1	5	**5 (Low)**	*Why it matters:* A single penny discrepancy in a multi-million-pound business triggers audit flags. *Mitigation:* 100% of arithmetic handled via `Decimal.js` and `NUMERIC(19,4)`. Banned JavaScript native `number` from financial pipelines.	Backend Lead	Controlled
RSK-02	**Accidental cross-tenant data leak** where Company A views Company B's client list or invoices.	1	5	**5 (Low)**	*Why it matters:* Catastrophic GDPR violation that would destroy market credibility. *Mitigation:* Composite foreign keys `(organization_id, id)` enforced at PostgreSQL schema level. The database engine rejects foreign associations crossing tenant lines.	Lead Architect	Controlled
RSK-03	**Retroactive modifications to historical periods** invalidating filed HMRC VAT returns.	1	5	**5 (Low)**	*Why it matters:* Can result in HMRC penalties and incorrect dividend distributions. *Mitigation:* Three-tier period locking state machine (`HARD_LOCKED`) physically blocks journal creation or validation within locked dates.	Product Owner	Controlled
RSK-04	**Concurrency race condition** generating duplicate	2	4	**8 (Medium)**	*Why it matters:* Breaks sequential ledger auditability mandated by Companies House. *Mitigation:* Atomic sequence increments	Backend Lead	Controlled

	journal sequence numbers during high traffic.				executed inside serializable PostgreSQL database transactions.		
RSK-05	**Changes in HMRC MTD API specifications** during Q1 2027 release window.	3	3	**9 (Medium)**	*Why it matters:* Could delay automated VAT filing launch. *Mitigation:* Isolated adapter service decoupling core ledger from HMRC transmission protocols.	Compliance Lead	Monitoring

2. Assumptions Log (Key Foundations)

- **ASM-01 (UK GAAP Alignment):** We assume target SME customers operate under standard UK GAAP (FRS 102 / FRS 105) reporting standards.
- **ASM-02 (Database Sovereignty):** We assume target UK clients require cloud infrastructure hosted within the UK / EEA (fulfilled by AWS `eu-west-2` London).
- **ASM-03 (Browser Compatibility):** We assume modern web client usage (Chromium, Safari, Firefox) with JavaScript enabled, supporting Next.js 15 App Router.

3. Issues Log (Resolved Technical Hurdles)

- **ISS-01 (Database Overlap Edge Cases):** Calendar calculations could accidentally generate overlapping financial periods during leap years.
- *Resolution:* Implemented PostgreSQL `btree_gist` temporal exclusion constraint, turning calendar overlaps into hard database errors.
- **ISS-02 (Audit Trail Vulnerability):** Concern that a malicious database administrator could execute SQL `DELETE` to hide ledger errors.
- *Resolution:* Added PostgreSQL database trigger `prevent_audit_log_mutation()` that intercepts and cancels any `UPDATE` or `DELETE` on `audit_logs`.

4. Dependencies Log (External Factors)

- **DEP-01 (PostgreSQL `btree_gist` Extension):** Requires hosting providers that permit custom PostgreSQL extensions (supported on AWS Aurora, Render, and Railway).
- **DEP-02 (HMRC Government Gateway API Access):** Production MTD filing requires registered HMRC Developer credentials and sandbox approval.